Group certificates for twenty-two hyperbolic table entries

Finite companion to the fixed-source recovery

Abstract

We identify actual automorphism groups of twenty-two marked quotients already certified as regular rank-four hypertopes. Explicit homomorphisms, normal kernels and complements establish the group descriptions in Table 3 of Fernandes–Leemans–Weiss. The proof uses no new normal-subgroup search. The two excluded orders and the ninth diagram remain separate obligations of the complete source contract.

1 The precise assertion

Put $E_d = \mathrm{C}_2^d$, $L_q = \mathrm{PSL}_2(q)$, and let X_{32} denote the actual extraspecial group of order32 certified below. A colon means a split extension, a cross a direct product, and a dot a nonsplit extension. An expression with an unspecified action denotes the actual extension supplied by its certificate; it is not a uniqueness assertion about every group with that order or series.

We acquire the complete eight-type quotient packet and its distinct review and root adoption. Its record (τ, j) is an ordered four-tuple of permutations; $H_{\tau,j}$ is the literal group generated by that tuple. The packet proves the exact Coxeter orders, all256 subset intersections and actual flag transitivity for every selected record. Thus these properties are not inferred here from the abstract group name. Its numbering is fixed; a freshly enumerated kernel list need not have the same order.

Theorem 1. *For each row of Table 1, the actual marked group $H_{\tau,j}$ has the displayed description. Consequently all twenty-two positive named entries of fixed-source Table 3 outside the diagram $(3,5,3,5)$ and the two excluded orders are realized by the previously certified hypertopes.*

The first four digits denote successive square edges; three digits denote a string. The label $5;3,3$ denotes the star with three edges at vertex1. The source's "locally spherical" diagram condition and the stronger universal-residue condition remain distinct, as in the acquired core. The source group notation need not uniquely fix an action. For these actual twenty-two records, the stored local orders equal the universal orders in the acquired core. Thus these witnesses also have universal rank-three residues; this follows from their exact local data, not their group names.

2 Certificate principles

For finite permutation groups $N, K \leq D$, the equations

$$N \lhd D, \qquad N \cap K = 1, \qquad \langle N, K \rangle = D$$

give $D = N \rtimes K$. A specified bijective homomorphism $K \to M$ identifies the complement with the named model M. An involution $t \notin D$ with $D \lhd \langle D, t \rangle$ and index2 gives another split step. If z is a central involution outside the preceding subgroup U, then $\langle U, z \rangle = U \times \langle z \rangle$. These statements follow immediately from the unique product expression in subgroups with trivial intersection; they require actual containment and normality, which the checker tests explicitly.

Diagram τ	Record j	Order	Certified description
$(3,3,3,4)$	8	1920	$E_4 : \mathrm{A}_5 : \mathrm{C}_2$
	9	2688	$(E_3 : L_7) \times \mathrm{C}_2$
	11	3840	$E_5 : \mathrm{A}_5 : \mathrm{C}_2$
	12	7680	$(\mathrm{C}_4 \,.\, (E_4 : \mathrm{A}_5)) : \mathrm{C}_2$
	13	7680	$E_6 : \mathrm{A}_5 : \mathrm{C}_2$
$(3,3,3,5)$	5	7200	$(\mathrm{A}_5 \times \mathrm{A}_5) : \mathrm{C}_2$
$(3,4,3,4)$	24	1152	$X_{32} : \mathrm{S}_3 : \mathrm{S}_3$
	27	1440	$\mathrm{S}_6 \times \mathrm{C}_2$
	36	4320	$\mathrm{S}_6 \times \mathrm{S}_3$
	38	5376	$(E_3 : L_7) \times \mathrm{C}_2^2$
	40	6048	$L_7 \times \mathrm{S}_3^2$
	45	9216	$E_8 : \mathrm{S}_3 : \mathrm{S}_3$
$[3,5,3]$	5	8160	$L_{16} : \mathrm{C}_2$
$[4,3,5]$	14	3840	$E_5 : \mathrm{A}_5 : \mathrm{C}_2$
	12	3840	$(E_4 : \mathrm{A}_5 : \mathrm{C}_2) \times \mathrm{C}_2$
	15	7680	$(E_5 : \mathrm{A}_5 : \mathrm{C}_2) \times \mathrm{C}_2$
	16	7680	$E_5 : \mathrm{A}_5 : \mathrm{C}_2 : \mathrm{C}_2$
$[5,3,5]$	5	7200	$(\mathrm{A}_5 \times \mathrm{A}_5) : \mathrm{C}_2$
	6	8160	$L_{16} : \mathrm{C}_2$
$[5;3,3]$	14	3840	$(E_5 : \mathrm{A}_5) \times \mathrm{C}_2$
	19	3840	$E_6 : \mathrm{A}_5$
	20	7680	$(E_6 : \mathrm{A}_5) \times \mathrm{C}_2$

Table 1: Actual tuples in the immutable eight-type core.

For the three $E_d : \mathrm{A}_5 : \mathrm{C}_2$ square rows the checker in fact constructs an S_5 complement. Its normal alternating subgroup and a transposition give the displayed successive split extensions. The two complements of order36 are explicitly $\mathrm{S}_3 \times \mathrm{S}_3$, so their displayed two-step series also follows, with the actual induced actions.

For X_{32} the checker finds $|X_{32}| = 32$ and $Z(X_{32}) = X_{32}'$ of order2, with generator squares in this central subgroup. The quotient is elementary abelian: its generators commute and have order at most2. Thus $X_{32}^2 X_{32}' = Z(X_{32})$; the elementary Frattini formula for a finite2-group shows that the Frattini subgroup also equals this order2 group. This proves the asserted extraspecial property; no sign or small-group identifier absent from the source is inferred.

For the affine rows we construct translations of $\mathbb{F}_2^3$ and the literal natural $\mathrm{GL}_3(2)$ action on its eight vectors. A retained bijective generator homomorphism identifies the linear group with L_7. The resulting affine group has order1344. The required direct products are then named models, compared by explicit bijective homomorphisms to the entire actual H. No group is identified merely by its order.

3 The central nonsplit row and execution interface

For $(\tau, j) = ((3,3,3,4), 12)$ the certificate gives

$$D \triangleleft H, \quad |D| = 3840, \quad [H : D] = 2, \quad Z = \mathrm{C}_4 \leq Z(D), \quad Z \cap D' \neq 1.$$

It also gives $N \triangleleft D$ of order64 containing Z, and a subgroup $U \leq D$ with a surjective homomorphism $U \to \mathrm{A}_5$ whose kernel is exactly Z. Generator squares and commutators of N lie in Z, so $N/Z \cong E_4$. The checked equalities $N \cap U = Z$ and $NU = D$ give $D/Z \cong E_4 : \mathrm{A}_5$. If the central extension by Z split, then $D = Z \times K$ for a complement K, and $D' = K'$ would be disjoint from Z. This contradicts $Z \cap D' \neq 1$, proving the dot assertion. An actual involution outside D generates H with D, proving the final split factor. The other split L_{16} rows additionally satisfy $C_H(D) = 1$ for their normal $D \cong L_{16}$; their external action is therefore faithful, rather than an unproved identification by order.

Finite verification. The three certificate files contain only literal permutations, specified subgroup generators, homomorphism images and outside elements. The verifier reads the original core tuple and reconstructs every group from generators. It checks containment, exact orders, normality, intersections, elementary abelian tests, each claimed full generated group, and actual bijectivity of each generator homomorphism. The twelve series records, nine direct-model records and one central-extension record have exactly the22 distinct keys in Table 1. Five damaged certificates independently corrupt a model image, an elementary kernel, an outside involution, the order4 central kernel, or the linear-group bridge; each is rejected.

These are direct positive certificates. Their earlier search routines are candidate providers only, and their execution is not a completeness premise. The accepted core separately supplies the necessary incidence mathematics and complete normal-quotient searches. Default replay checks all22 groups and the five damaged inputs; it does not rerun those searches. Python optimization is rejected before output. Complete native sources are included.

Sources, software and scope. The source is M. Fernandes, D. Leemans and A. I. Weiss, the fixed sixteen-page L033 article, Table 3 on physical page14. Its positive group entries are the objects recovered here; we make no priority claim. The exact permutation, homomorphism and finite-field calculations use GAP4.15.1. Its documented $\mathrm{PSL}(2, q)$ constructor is the action of determinant-one matrices on projective lines, and the standard alternating/symmetric constructors are named-model interfaces. These software interfaces are retained explicitly, not formally verified by this computation. The affine model itself is built from the stated vector action. The acquired LINS completeness and finite classification premises are unchanged and are not used for a new census here.

The project root supplied the original normal-quotient probe; the publication agent authored these group certificates, verifier and companion proof. Earlier distinct core reviews are acquired with their original roles. The two source orders336 and2160 are excluded by that core, not by this naming argument. The ninth diagram, its two printed orders3840 and9720, and all remaining whole-source obligations are outside this theorem.